

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455519.5899 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.018
Transit depth (Rp/Rs)^2	2.02 +/- 0.5 [%]
Orbital Inclination (inc)	84.91 +/- 1.11
Airmass coefficient 1 (a1)	1.215 +/- 0.013
Airmass coefficient 2 (a2)	-0.1097 +/- 0.0079
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	None
Optimal Aperture	3.66
Optimal Annulus	7.96
Transit Duration (day)	0.074 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.018 vs 0.1326 (deviation 0.52σ)
Duration fit vs published	0.074 d vs 0.0737 d (deviation 0.02σ)
Mid-time O–C	-4.5 min
Depth SNR	7.9
Residual scatter	1.07 %

Light curve

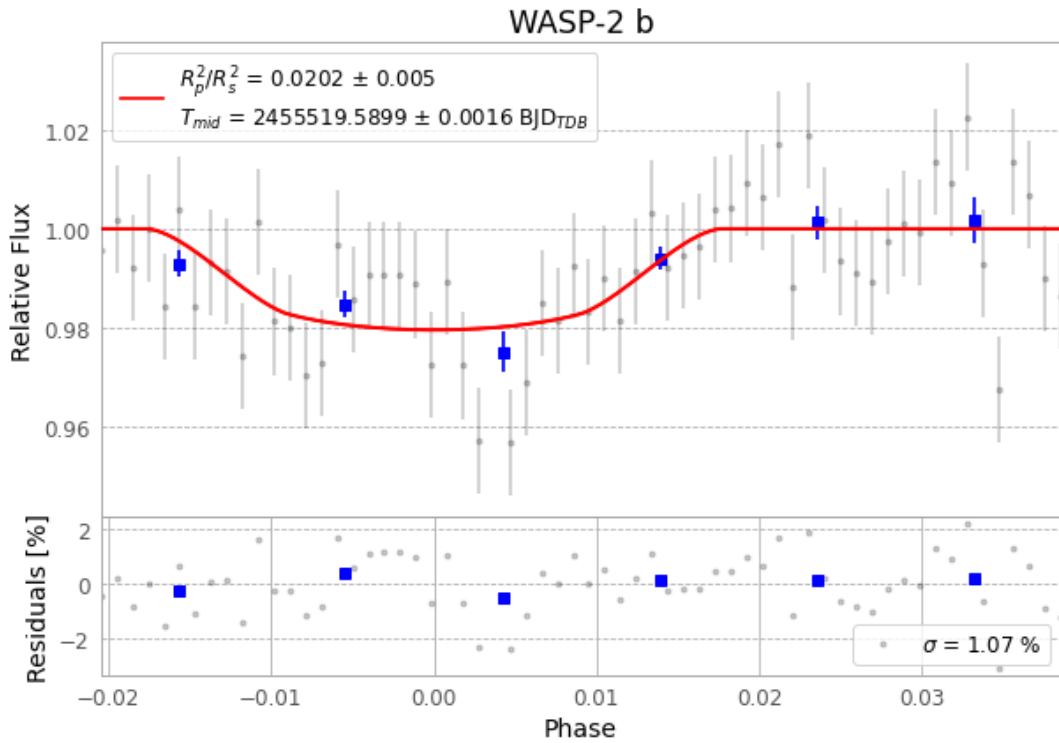


Figure 1 — FinalLightCurve_WASP-2 b_2010-11-18.png (SHA-256 78924cdd8276d754...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [203, 204], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 977d5f3503c81219...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

62 FITS frames (41 MB), WASP-2101119010614.FITS → WASP-2101119040913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-101119.log (aa5f365a5cbb...), FinalLightCurve_WASP-2 b_2010-11-18.png (78924cdd8276...), FinalLightCurve_WASP-2 b_2010-11-18.csv (443abe6cbb28...)

Compute resources

Wall time 91.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:19Z.